MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE
DIODE, SILICON, POWER RECTIFIER
TYPE - 1N2153

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a silicon power rectifier with the following characteristics:

Ratings	Vf Vdc	Vac rms	if A	IF Adc	(i (surge)) 1-120 sec	T stg °C
Absolute Maximum	600	420	6	5	50	+ 150
Minimum	Note 1 ---	Note 1 ---	Note 2 ---	--- ---	--- ---	--- -65

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and publications, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATION

MILITARY

MIL-S-19500 Semiconductor Devices, General Specification for.

(Copies of specifications are required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements.- Requirements shall be in accordance with Specification MIL-S-19500, and as specified herein.

Source: https://assist.dla.mil -- Downloaded: 202610-18T13:10Z
Check the source to verify that this is the currentversion before use.

3.2 Qualification.- Qualification shall be as specified in 4.3.

3.3 Definitions.- Definitions shall be as defined in MIL-S-19500.

3.4 Abbreviation and symbols.- Abbreviations and symbols shall be in accordance with MIL-S-19500 and as specified herein:

hr----------hour
mm ----------millimeter

3.5 Design and construction.- Diodes shall be of the design, construction, and physical dimensions specified on figure 1.

3.5.1 Operating position.- Diodes shall operate in any position.

3.5.2 Maximum operating altitude.- Diodes shall operate at a maximum-altitude equivalent to 15 mm of mercury.

3.6 Performance characteristics.- Performance characteristics shall be as specified in tables I, II, and III.

3.7 Marking.- Diodes conforming to all the requirements of this specification and passing Government inspection shall be marked "U.S. Army, qualification approval code letters, and the number IN2153. If a waiver has been granted, only the commercial brand shall be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor responsibility.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Inspection conditions and methods of test.- Inspection conditions and methods of test shall be in accordance with Specification MIL-S-19500 appendix C, and as specified herein.

4.3 Qualification inspection.- Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III. The number of specimens to be inspected in each group, the determination of defectives, and the number of defectives will be as specified in the qualification-inspection routine in Specification MIL-S-19500. In table II the tests specified as end points will be performed after each test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:10Z
Check the source to verify that this is the current version before use.

4.4 Group A inspection.- Group A inspection shall consist of the examinations and tests specified in table I.

Table I Group A Inspection (also Group I Qualification)

Ref par	Examination or test	Conditions	AQL (percent defective)	Inspection Level	Symbol	Limits Min	Limits Max	Unit
	Subgroup 1							
30.13	Visual and mechanical examination		1.0	II	--	--	--	--
	Subgroup 2							
60.1.4	Forward voltage	if = 6 A_{dc} Note 6 Note 17			V_F	--	1.5	V_{dc}
60.1.10	Reverse current	Vdc = 600 Note 5			I_R	--	50	uAdc
30.6	High temperature Operating Test:							
60.1.10	Reverse Current Avg	T-case-150°C +3°C -0°C Vac = 420 f = cps I_F = 5A Note 9	2.5	II	I_R	--	750	uAdc
60.1.4	Forward Voltage	T-case = 150 I_F = 5 A_{dc} Note 6			V_F	--	2	Vdc
60.1.7	Peak Reverse voltage	I_R = 1.0 mAdc Temp. = 25° ±3°C Note 5			Vr	600	--	Vdc
60.1.15	Surge current	I_F = 5 Adc Vac = 420 f = 60 cps			if (surge)	--	50	A

Source: https://assist.dla.mil -- Downloaded: 202610-18T13:10Z
Check the source to verify that this is the current version before use.

4.5 Group B inspection.- Group B inspection shall consist of the tests specified in table II, in the order shown in each subgroup. Readings in addition to the end points specified in table II may be taken at the discretion of the manufacturer.

Table II. Group B inspection

Ref Par	Examination or test	Conditions	AQL (percent defective)	Inspection Level Code	Sym	Limits Min	Limits Max	Unit
	Subgroup 1 (also group II Qualification)							
30.9	Physical dimensions		4.0	L-7 Reduced L5 Procedure R1				
	Subgroup 2 (also group III Qualification							
40.13	Soldering							
40.6	Moisture-resistance	Notes 3, 10				10		cycle
40.14	Temperature cycling	Notes 3, 11		L-4 Reduced 12		5		cycle
40.16	Thermal shock	Notes 3, 18				5		
	End Points		4.0					
60.1.4	Forward voltage	if = 6 A_{dc} Note 6 Note 17		V_F				
60.1.10	Reverse current	V_{dc} = 600 Note 5			I_R	--	75	uAdc
	Subgroup 3 (also group IV Qualification)							
40.10	Shock	Notes 3, 12						
40.18	Vibration fatigue	Notes 3, 14						
40.4	Centrifuge	Notes 3, 15	4.0	L4	--	5000	--	G's
	End Points			Reduced L2 Procedure R1				
60.1.4	Forward voltage	if = 6 A_{dc} Note 6 Note 17			V_F	--	1.5	V_{dc}
60.1.10	Reverse current	V_{dc} = 600 Note 5			I_R	--	75	uAdc

Source: https://assist.dla.mil -- Downloaded: 202610-18T13:10Z
Check the source to verify that this is the current version before use.

Table II. Group B inspection (cont'd)

Ref Par	Examination or test	Conditions	AQL (percent defective)	Inspection Level Code	Sym	Limits Min	Limits Max	Unit
	Subgroup 4 (also group V Qualification)							
	Insulation resistance	V_{dc} = 600 Note 2				$10x10^5$		Meg
	Terminal tension	Note 1						
	End points		4.0	L4 Reduced L2 Procedure R1				
60.1.4	Forward voltage	if = 6 A_{dc} Note 6 Note 17			V_F	--	75	uAdc
60.1.10	Reverse current	V_{dc} = 600 Note 5			I_R	--	75	uAdc
	Subgroup 5 (also group VI Qualification)							
30.7	Low temperature operation:	T = -65°±3°C	4.0	L4 Reduced L2 Procudure R1				
60.1.7	Peak reverse voltage	I_R = 1.0 mAdc			Vr	600	--	V_{dc}

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:10Z
Check the source to verify that this is the current version before use.

may be waived by the contracting/acquisition activity upon certification by

4.6 Group C inspection. - Group C inspection shall consist of the tests specified in Table III, in the order shown for each subgroup. In addition to the specified end points, the manufacturer at his discretion may take other readings.

Table III. Group C inspection

Ref par	Examination or test	Condition	AQL (percent defective) 1,000 hour life test: 250 hrs	500 hrs	1,000 hrs	reduced hours life test: 500 hrs	250 hrs	Inspection level code 1,000 hour life test	reduced hours life test: 500 hrs	250 hrs	Sym	Limits Min	Max	Unit
	Subgroup I (also group VI Qualification)													
40.7.1.1	Operation life	+150° ±3°C t = 1000 hrs f = 60 cps Vac = 420												
	End points:	I_F = 5 A_{dc}												
	Forward voltage	if = 6 A_{dc} Note 6 Note 17	6.5	6.5	6.5	2.5	1.0	L6	L6	L6	V_F		1.5	V_{dc}
	Reverse current	V_{dc} = 600 Note 5									I_R		75	uAdc
	Subgroup II (also group VII Qualification)													
40.7.1.1	Storage life	Tstg = 150° ±3°C t = 1000 hours	6.5	6.5	6.5	2.5	1.0							
	End points:													
	Forward voltage	if = 6 A_{dc} Note 6 Note 17						I	I	I	V_F		1.5	Vdc
	Reverse current	V_{dc} = 600 Note 5									I_R		75	uAdc

Source: https://assist.dla.mil -- Downloaded: 202610-18T13:10Z
Check the source to verify that this is the current version before use.

Note 1. Terminal tension test. This test is to determine that the ceramic insulating disc is properly secured to the stud and soldering lug. The device shall be securely fastened by screwing the stud into a suitable support so that the terminal (cathode) extends downward in a vertical position. With a bandvise, or equivalent, the specified weight of 10 pounds avoirdupois (including the attaching device) shall be suspended from the terminal for a period of 20 seconds minimum.

Note 2. Insulation resistance. The electrical insulation shall be measured by using a megohmeter, the diode rectifier shall be mounted on an aluminum beat sink 6" x 6" x 1/4" and applying the specified voltage between the cathode and the beat sink.

Note 3. At the conclusion of this test the unit shall be tested in accordance with the conditions specified for each subgroup and point requirement.

Note 4. The peak reverse voltage shall be measured using the circuit in figure 1 or an equivalent circuit.

Note 5. With the specified dc voltage applied to the electrododes, the reverse current of the semiconductor device shall be measured. This test may be made utilizing volt-ammeter methods or by oscilloscope presentation with a sweep frequency of 10,000 cps or less.

Note 6. With the specified current flowing through the crystal diode, the voltage drop across its terminals shall be measured. This test may be made utilizing volt-ammeter methods or by oscilloscope presentation with a sweep frequency of 10,000 cps or less.

Note 7. The crystal diode rectifier shall be subjected to continuous operation life test in accordance with the following procedure:

(a) Insert the crystal in a half wave rectifier circuit.

(b) The applied signal amplitude, frequency and wave shape, the value of the load resistor and the duration of the test shall be specified on the specification.

Source: https://assist.dla.mil -- Downloaded: 202610-18T13:10Z
Check the source to verify that this is the current version before use.

(c) Specified measurements to determine compliance with the specified life test end points shall be made at the end of the first twenty-four hours and at intervals thereafter as specified until the completion of the life test. The duration of the life test shall be as specified in table III.

Note 8. The criteria for life test end point shall be as specified on table III. A semiconductor device shall have reached the end of its life when it fails the specified life test end point limit(s) when measured under the specified test conditions. When two or more tests are specified for life test end points, failure of any one of these tests shall constitute failure of the semiconductor device.

Note 9. High temperature operating test. With the specified voltages applied, the semiconductor device shall be operated at 150°C temperature. After thermal equilibrium has been reached at the specified high temperature, performance measurements shall be made, using the test circuit shown on figure 2.

High temperature operating test circuit

The reverse leakage current of each crystal rectifier, D_1 & D_2 must be less than 1/20 of the maximum allowable specified leakage of the rectifier under test. In other respects, crystal rectifiers D_1 & D_2 should be the same, type as the rectifier under test.

Note 10. The semiconductor device shall be tested in accordance with Method 106 per Standard MIL-STD-202.

Note 11. The semiconductor device shall be subjected to the number of temperature cycles as specified. Low temperature shall be -65°C or lower; high temperature shall be 150°C or higher. The device shall be maintained at each end temperature for sufficient time to reach thermal equilibrium, but for not less than 15 minutes. The time for changing the temperature from end point to end point shall not exceed 30 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:10Z
Check the source to verify that this is the current version before use.

Note 12. The diode unit shall be tested in accordance with Method 202, Drawing 254-JAN. The diode unit shall be subjected to three 500 G shocks of approximately 1 millisecond duration in each of two directions, one direction along the axis, and the other perpendicular to the axis.

Note 13. The semiconductor device shall be fastened rigidly on the vibration platform. The device shall be vibrated successively in two directions (one along the axis and the other perpendicular to the axis of the diode unit) over a frequency range of 100 to 1000 cycles per second, covering the vibration frequency range four times in no less than 16 minutes, with uniform variation of frequency and constant peak acceleration of 10G.

Note 14. The semiconductor device shall be fastened rigidly on the vibration platform and shall be vibrated with simple harmonic motion at any singly frequency between 45 and 100 cps, with a constant peak acceleration of 10G. The diode shall be vibrated for a total of 64 hours, 32 hours in each of the two directions, one along the axis and the other perpendicular to the axis of the diode.

Note 15. The diode units shall be subjected to a centrifugal acceleration of 5000 G's in each of two directions (one along the axis and the other perpendicular to the axis of the diode unit). The centrifuge shall be held at the maximum speed until a stable reading of the tachometer is obtained.

Note 16. Measurements made with the unit mounted on a 6" x 6" x 1/4" beat sink.

Note 17. Absolute maximum ratings. See definition of absolute maximum ratings given in paragraph 20.1 of MIL-S-19500. Absolute maxiumum ratings (excluding storage temperature) are for T case at 150°C.

Note 18. The diode shall be immersed in a liquid at 100°C ± 5°C for a minimum of 5 minutes. The diode shall then be transferred to a liquid of 0° ± 5°C, transfer time shall be less than 10 seconds. The diode shall remain at 0°C for 5 minutes then similarly transferred to the liquid at 100°C. This constitutes one complete thermal shock cycle. At the end of the number of thermal shock cycles specified there shall be no evidence of mechanical damage to the diode.

5. PREPARATION FOR DELIVERY

5.1 Semiconductor devices shall be prepared for delivery in accordance with Specification MIL-S-19500. In addition, semiconductor devices which may be damaged by electromagnetic radiation shall be individually and completely enclosed in a nonmagnetic electrical conducting covering for electrical shielding.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Title, number, and date of the applicable detail specifications, and the complete type designation.

(c) Level of preservation, and packaging, and packing and applicable marking.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:10Z
Check the source to verify that this is the current version before use.

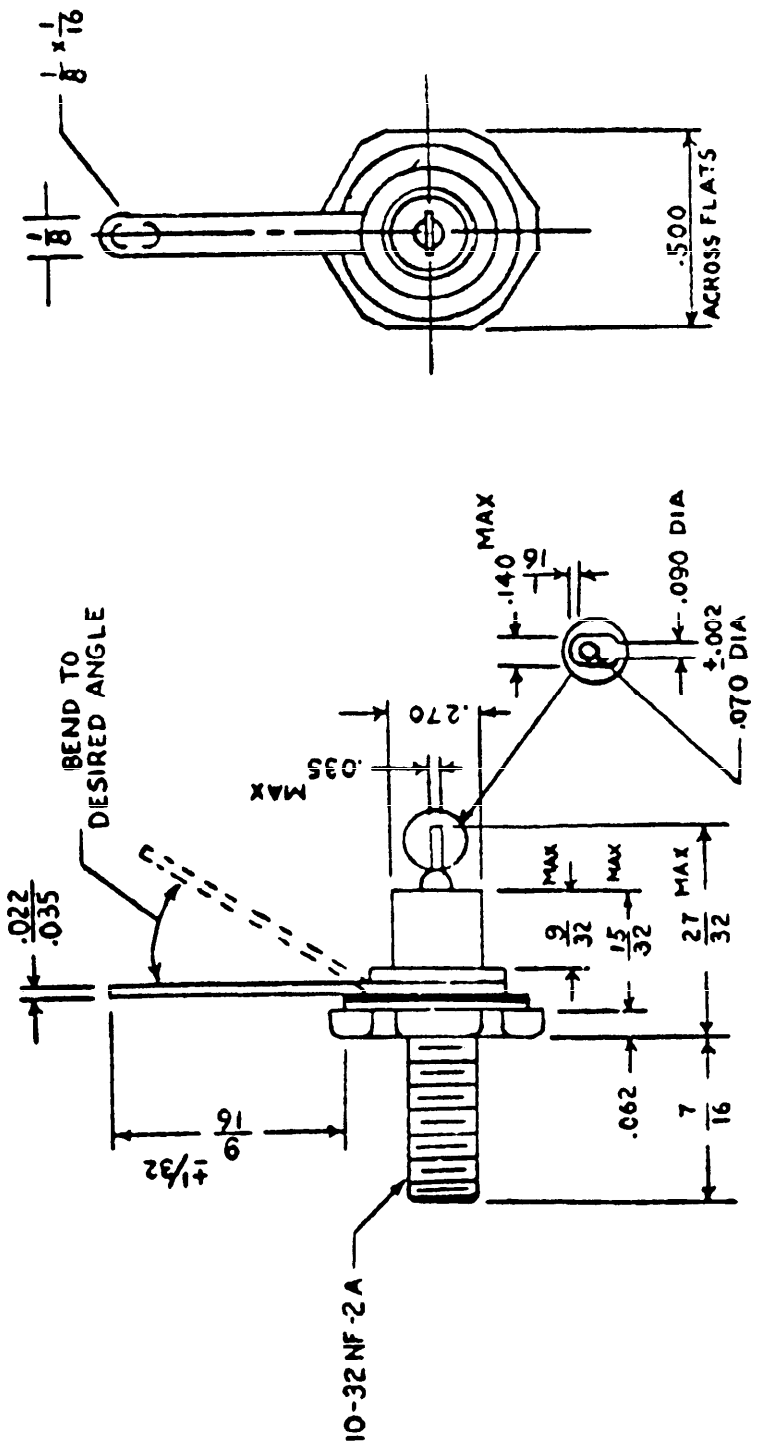
6.2 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids been tested and approved for inclusion in the applicable Qualified Products List QPL-19500 Army whether or not such products have actually been so listed by that date.

6.2.1 The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the U. S. Army to be tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Application for qualification approval should be addressed to Components and Materials Branch, Standardization Division, U. S. Army Signal Equipment Support Agency, Fort Monmouth, New Jersey.

NOTICE - When Government drawings, specifications, or other data are used for any other purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specification, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 202610-18T13:10Z
Check the source to verify that this is the currentversion before use.

SILICON POWER RECTIFIER - 1N2153



Source: https://assist.dla.mil -- Downloaded: 202610-18T13:10Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 202610-18T13:10Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDING GENERAL
U.S. ARMY ELECTRONICS COMMAND
ATTN: AMSEL-PP-E-SED
FORT MONMOUTH, N.J. 07703

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:10Z
Check the source to verify that this is the current version before use.